

MARKETMINT

PRESENTATION



A Scalable Full- Stack E-Commerce System

Built with the modern technologies and deployed on AWS for enterprise grade performance and reliability

OVERVIEW

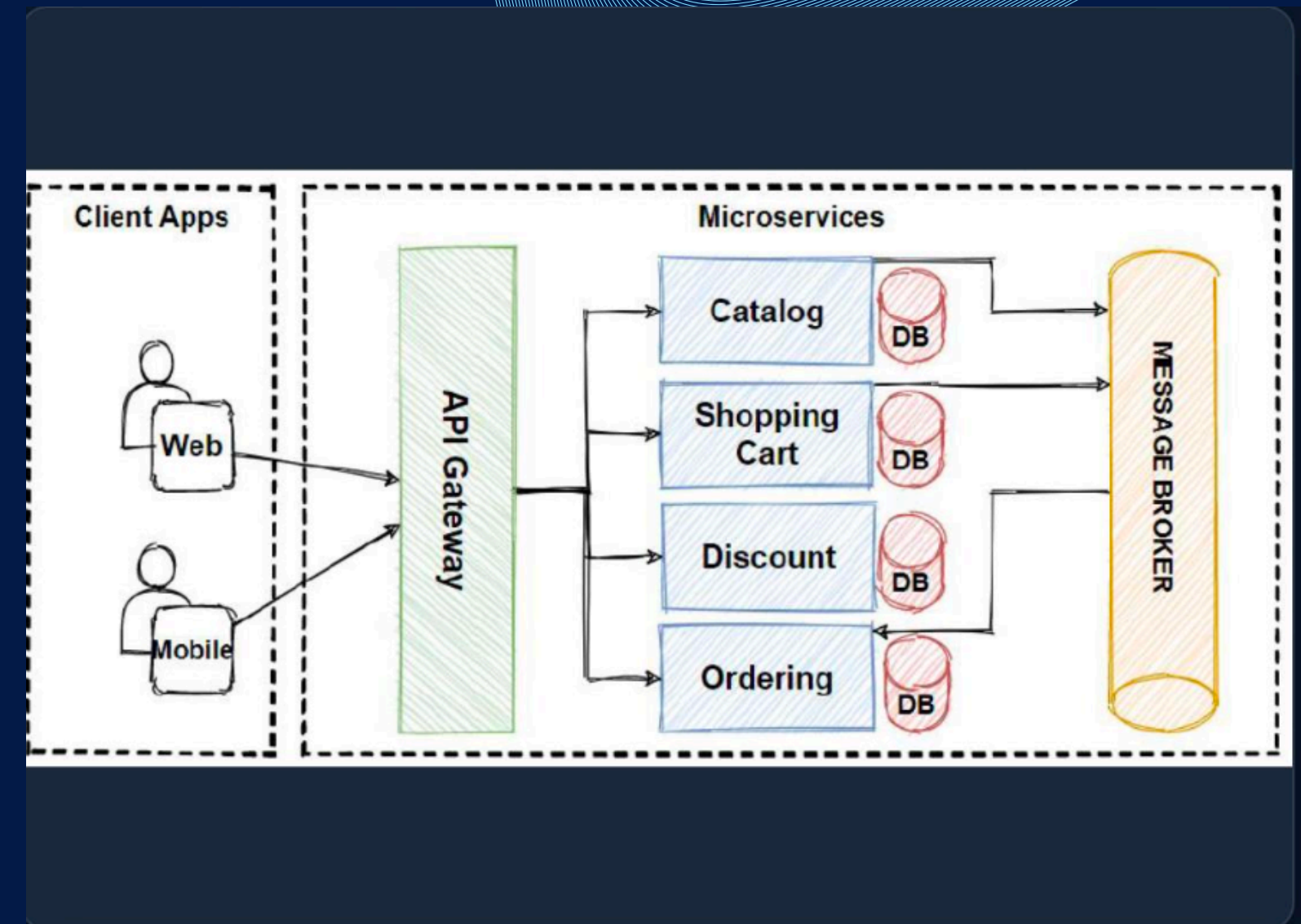
AGENDA

- Project Architecture
- Services
- Frontend and Backend Implementation
- AWS setup
- Performance
- Challenges
- Demo

ARCHITECTURE

Our architecture ensures secure data flow and high availability through a multi-tier cloud deployment.

- **Frontend:** EJS templating with dynamic HTML, CSS, and JS.
- **Backend:** Robust Nodejs runtime with Express framework.
- **Database:** Non-relational MongoDB for flexible data schemas.
- **AWS Cloud:** EC2 computing, S3 storage, and CloudFront CDN
- **Gateway:** Nginx Reverse Proxy for load balancing.



AWS CLOUD SERVICES

AWS SES-
EMAIL SENDING



AWS S3-
STATIC
CONTENT
STORAGE



AWS EC2-
BACKEND
DEPLOYMENT



AWS
CLOUDFRONT-
GLOBAL CDN



Frontend Experience

Dynamic User Interface

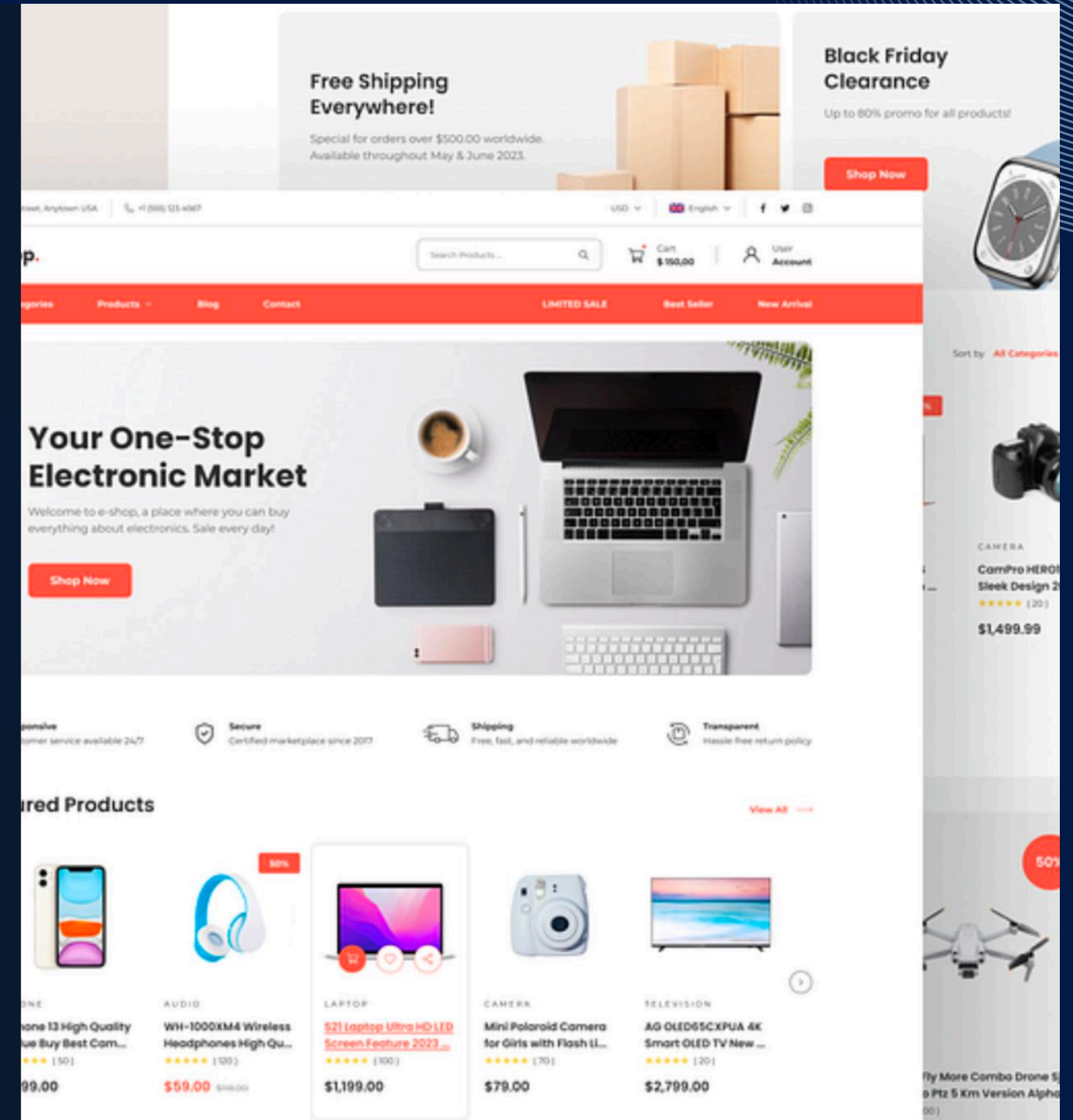
We utilized **EJS templates** to render dynamic content server-side, ensuring fast initial page loads and SEO friendliness.

Responsive Design

Custom CSS styling provides a seamless shopping experience across devices, with an intuitive layout mimicking industry leaders.

Asset Optimization

All static assets (images, stylesheets) are delivered via **AWS S3 and CloudFront**, reducing latency for global users.



Backend Engineering



Core Logic

Built on **Node.js & Express**. Structured with MVC architecture separating Routes, Controllers, and Models for maintainability.



Data Layer

Integrated **MongoDB with Mongoose** ORM. Implements complex schemas for Users, Products, and Orders with validation.



Cloud Integration

AWS SES handles transactional emails. **EC2** hosts the application logic, while **S3** manages scalable file storage.

AWS SETUP

Performance Metrics

<200ms

Average API Response

Optimized for Speed

Global CDN: CloudFront caches content at the edge, drastically reducing latency for static assets.

Process Management: PM2 keeps the Node.js application alive and utilizes clustering for load distribution.

Database Indexing: Mongoose queries are optimized to handle inventory searches efficiently.

Technical Challenges



SES Sandbox Limitations

Overcame AWS email sending restrictions during the development phase by verifying specific test emails.



Nginx Configuration

Configuring the reverse proxy to correctly forward headers and handle SSL termination for the EC2 instance.



S3 & CORS Policies

Debugging Cross-Origin Resource Sharing permissions to allow the frontend to securely fetch images from S3 buckets.



Schema Design

Designing a flexible yet strict MongoDB schema to handle product variations and order history.

DEMO



THANK YOU